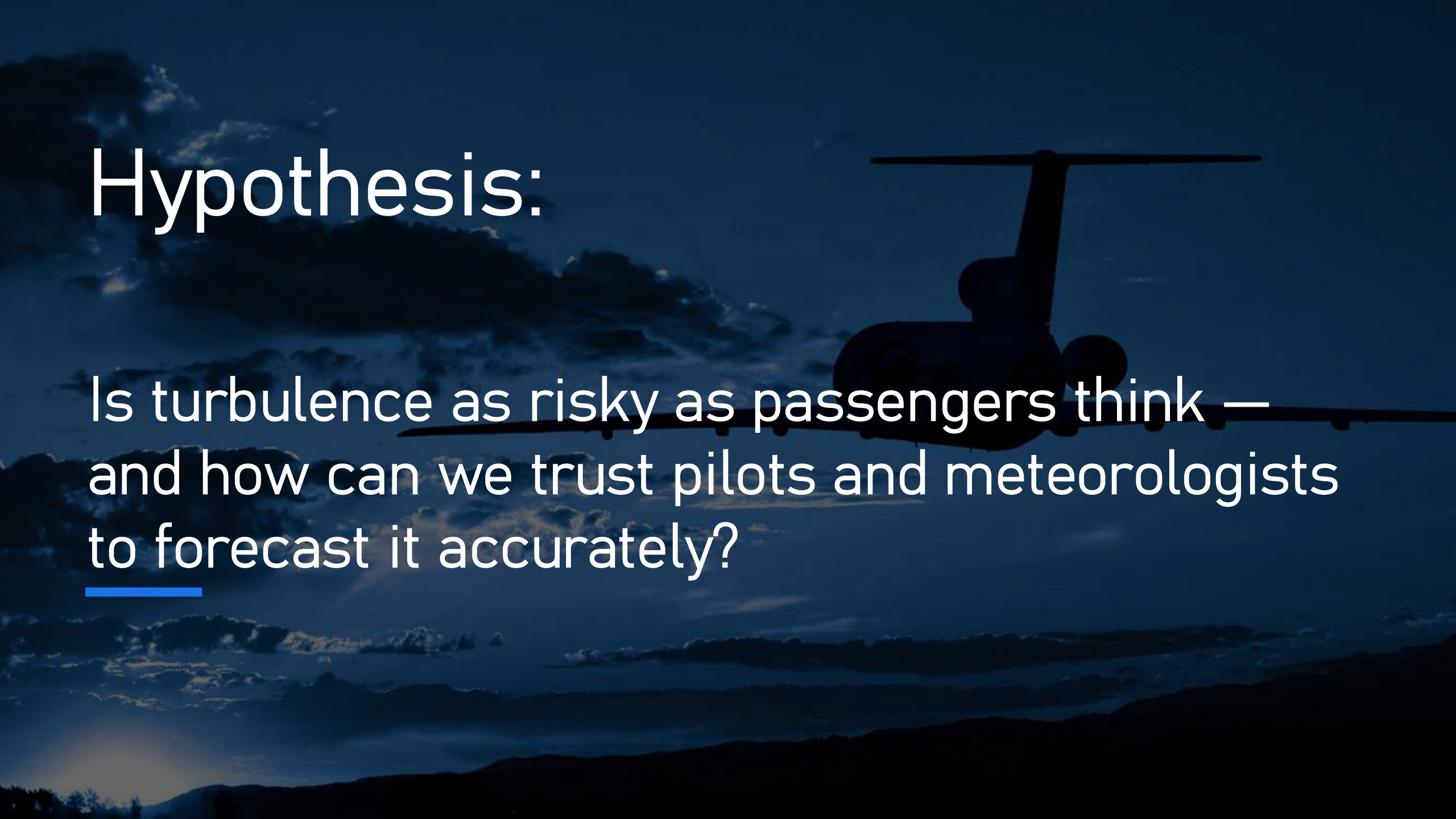


How do passengers perceive turbulence risk compared to actual forecast accuracy?

BY CECE WEAVER

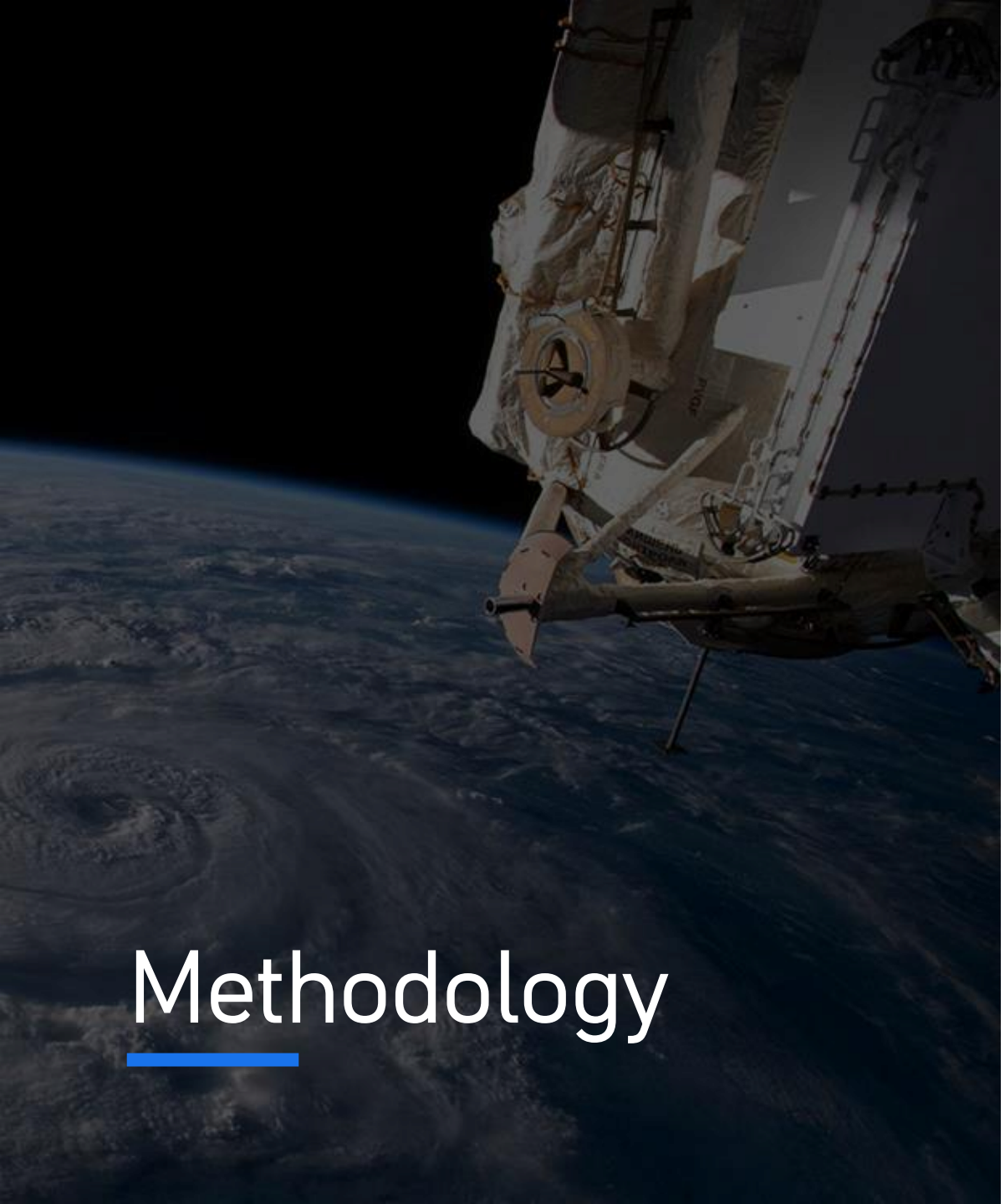


Hypothesis:

Is turbulence as risky as passengers think — and how can we trust pilots and meteorologists to forecast it accurately?

Secondary Research

- Meteorologists use a lot of equations to explain it, but it's unpredictable
- Climate change is increasing turbulence globally
  (Williams, 2017)
- ➢ Forecasting technologies like ECMWF IFS & Doppler spectrum increase turbulence predictability
- Turbulence has not caused a fatal crash in over 40 years
  (Insider Tech, 2022)
- Turbulence is the leading cause of anxiety among airline passengers
- Pilots are trained on turbulence protocol, but communication is lacking



Methodology

- 106 participants

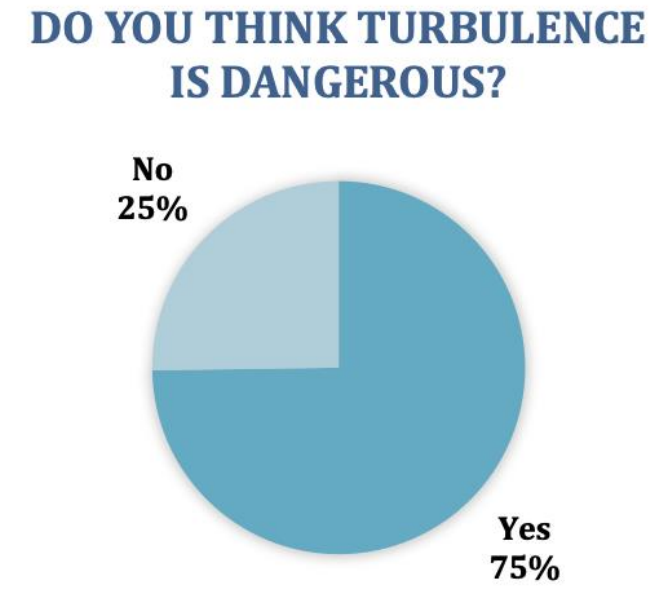
via SurveyMonkey

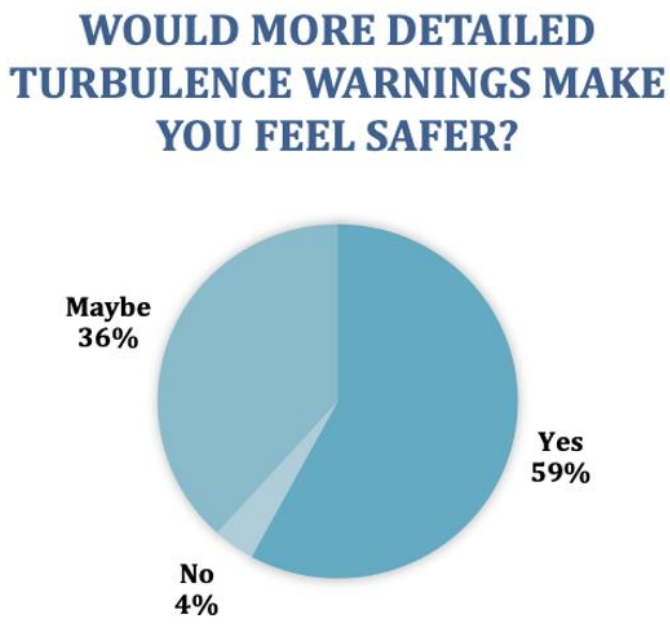
- Questions focused on turbulence experience, fear levels, and trust in airline industry

- Forecasting data review:
  - AIRMETS
  - PIREPs
  - ECMWF forecast maps & data models
  - Case studies of Denver-based flights

Results:

- **74.77%** of respondents said turbulence felt "**dangerous**"
- **59%** wanted more detailed **in-flight warnings**
  - Most were **not aware** of modern forecasting tools
  - Many **avoided flying near** mountains or during storms





Discussion:

Fear of Turbulence comes from:

Lack of in-flight communication	*Psychological discomfort*

- Most passengers misunderstand turbulence mechanics
- Pilots use weather tools to avoid turbulence, but some of the best data is locked behind private systems that not all airlines can access. *(Staigle, 2018)*
- Better communication = more trust in aviation safety

Next Steps:

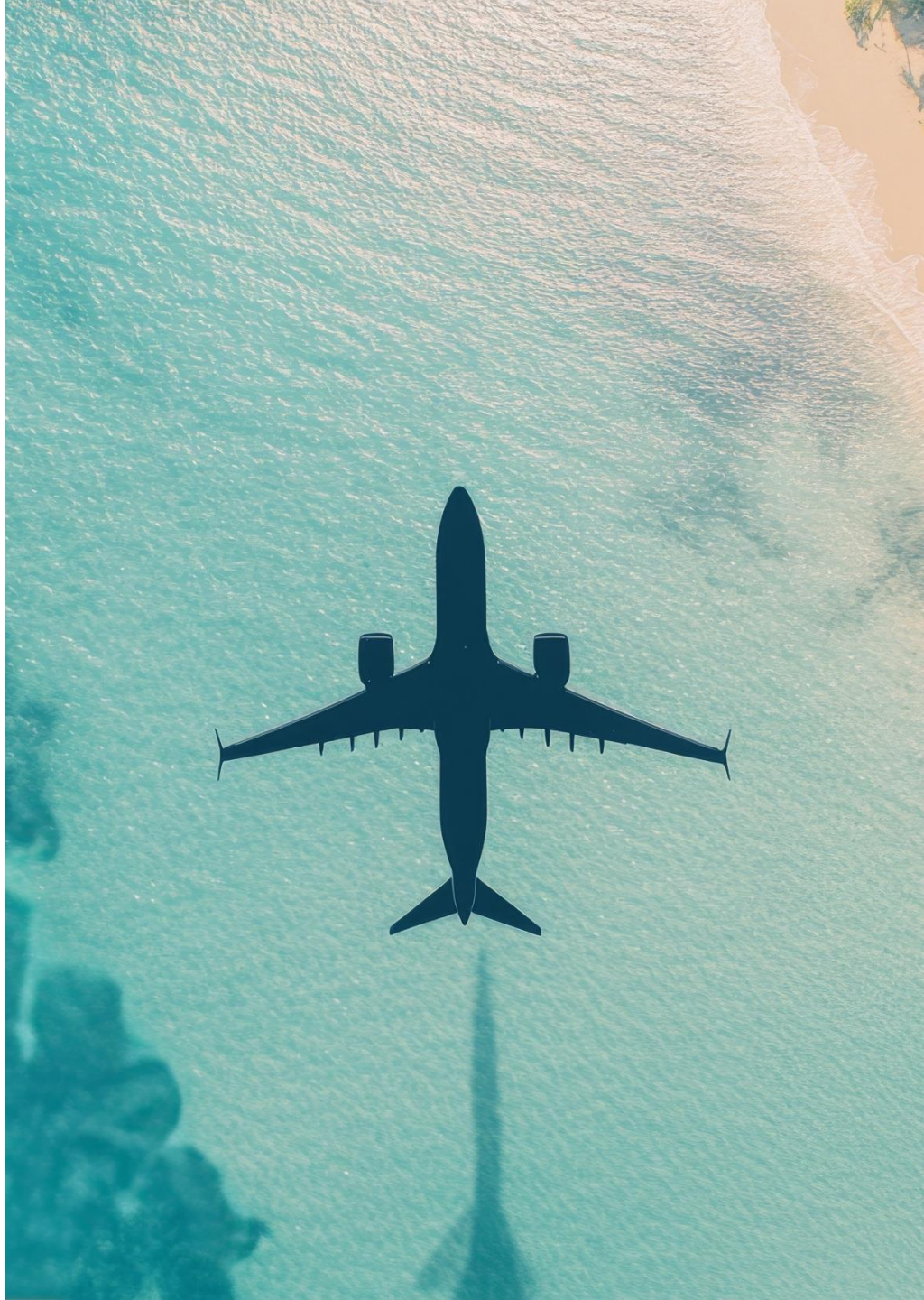
More public education about forecasting is needed

Airlines could reduce fear by:

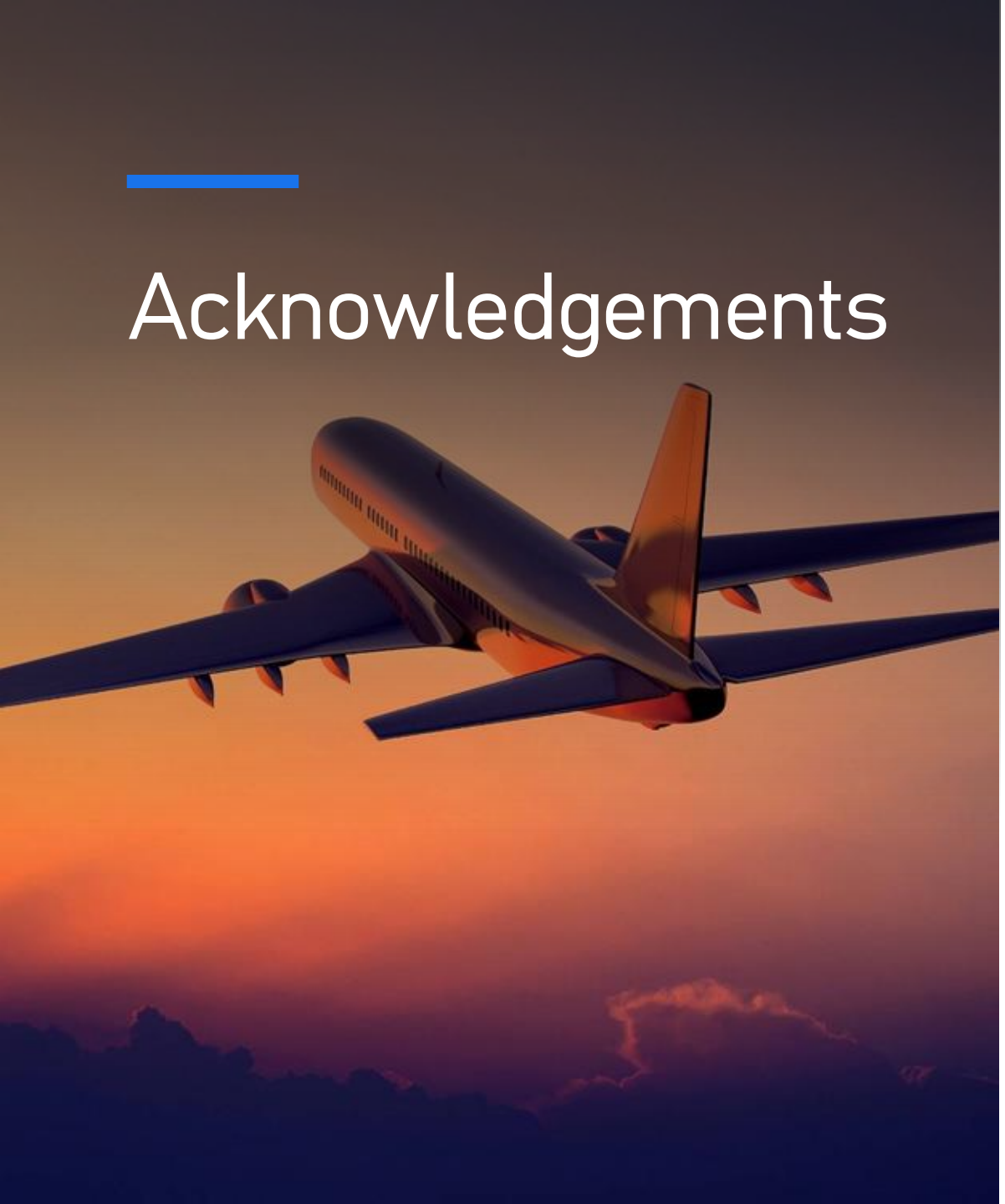
- Training crews in clear communication
- Offering simple turbulence explanations during flights

Future research would:

- Survey passengers in mountainous regions
- Compare perception vs. forecast data in real time



Acknowledgements



Professor Matthews and Peers

Survey participants

NOAA and FAA public data

Delta Airlines' turbulence safety protocols
(Staigle, 2018)


Thank you!